

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklists:\

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the Python, HTML, and Flask code.	Yes	Improves readability and maintains coding standards.
2	Proper File Structure	Project files and folders are organized into dataset, model, templates, static, and application files	Yes	Easy to navigate and maintain.
3	Meaningful Variable Names	Variables are named based on their purpose (e.g., life_expectancy, gni_per_capita, predicted_hdi).	Yes	Makes the code easy to understand.
4	Function / Method Names	Functions have descriptive names (e.g., predict_hdi(), load_model(), preprocess_data()).	Yes	Clearly indicates each function's purpose.
5	Code Comments	Important logic and functions are documented using comments.	Yes	Helps future developers understand the code.
6	Modular Design	The application is divided into reusable modules such as data preprocessing, model training, prediction, and Flask interface.	Yes	Promotes code reuse and easier maintenance.
7	No Redundant Code	Duplicate and unused code is removed to improve efficiency.	Yes	Reduces complexity and improves performance.
8	Error Handling	Input validation and exception handling are implemented to prevent application failures.	Yes	Ensures robust and reliable execution.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Data cleaning and preprocessing	High
2	Machine Learning Model	Scikit-learn	HDI prediction	High
3	Model Loading Module	Joblib	Flask application	High
4	Prediction Function	Python	User input prediction	High
5	Flask Web Interface	Flask, HTML, CSS	User interaction and prediction display	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project follows a clean and organized folder structure.
Readability	5	Meaningful variable names, proper formatting, and comments improve readability.
Reusability	5	Modular design allows components to be reused across different parts of the project.
Documentation / Comments	4	Adequate comments are included to explain key logic and functions.
Overall Score	4.8 / 5	The project follows good coding practices, is maintainable, reusable, and easy to understand.